Functional Analysis

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1. Metric spaces

Definition 1.1 (Metric space)**.** Let X be a set. A function $d\colon X \times X \to \mathbb{R}$ is a *metric* if for all $x, y, z \in X$ it satisfies $d(x,y) \geq 0$ with equality exactly when $x = y$, the symmetry $d(x,y) = d(y,x)$, and the triangle inequality $d(x,z) \leq d(x,y) + d(y,z)$. The pair (X,d) is a *metric space*.

Definition 1.2 (Cauchy sequence)**.** A sequence (x_n) in a metric space (X,d) is *Cauchy* if for every $\varepsilon > 0$ there is an $N \in \mathbb{N}$ such that $d(x_n, x_m) < \varepsilon$ for all $n, m > N$.

Definition 1.3 (Completeness)**.** A metric space (X,d) is *complete* if every Cauchy sequence in X converges to a limit lying in X.

Example 1.4 (Shortest-path metric)**.** Let $G = (V,E)$ be a connected graph and let $d(v_1, v_2)$ be the length of a shortest path between v_1 and v_2. Then d is a metric on V. Connectedness is what makes d finite.

Proposition 1.5 (Completion)**.** For every metric space (X,d) there is a complete metric space $(\overline{X}, \overline{d})$ together with a map $i\colon X \to \overline{X}$ such that $\overline{d}(i(x), i(y)) = d(x,y)$ for all $x, y \in X$ and $i(X)$ is dense in $\overline{X}$. It is unique up to isometric isomorphism: any two such spaces are related by a bijective isometry carrying one copy of X onto the other. $(\overline{X}, \overline{d})$ is the *completion* of (X,d).

Example 1.6. The space $(\mathbb{Q}, |\cdot|)$ is not complete; its completion is $(\mathbb{R}, |\cdot|)$.

Remark 1.7. It is common to suppress i and assume outright that $X \subset \overline{X}$, in which case the last condition reads: X is dense in $\overline{X}$.

2. Normed spaces and the p-norms

Definition 2.1 (Norm)**.** Let X be a vector space over $\mathbb{K}$, where $\mathbb{K}$ is $\mathbb{R}$ or $\mathbb{C}$. A function $\|\cdot\|\colon X \to \mathbb{R}$ is a *norm* if for all $x, y \in X$ and $\lambda \in \mathbb{K}$ it satisfies $\|x\| \geq 0$ with equality exactly when $x = 0$, the homogeneity $\|\lambda x\| = |\lambda| \|x\|$, and the triangle inequality $\|x + y\| \leq \|x\| + \|y\|$. Then $(X, \|\cdot\|)$ is a *normed space*, and $d(x,y) = \|x - y\|$ is a metric on X.

Example 2.2.

1. $X = \mathbb{K}^n$ with the p-norm $\|x\|_p = (\sum_{j=1}^{n} |x_j|^p)^{1/p}$ for $1 \leq p < \infty$, or with $\|x\|_\infty = \max_{1 \leq j \leq n} |x_j|$.
2. $X = C[0,1]$ with the supremum norm $\|f\|_\infty = \max_{t \in [0,1]} |f(t)|$, or with the p-norm $\|f\|_p = (\int_0^1 |f|^p)^{1/p}$.
3. $X = C^1[0,1]$ with $\|f\| = \max_t |f(t)| + \max_t |f'(t)|$.
4. $\ell_p = \left\{ x = (x_n) \;\middle|\; \sum_{n=1}^{\infty} |x_n|^p < \infty \right\}$ with $\|x\|_p = (\sum_{n=1}^{\infty} |x_n|^p)^{1/p}$.
5. ℓ_∞, the bounded sequences, with $\|x\|_\infty = \sup_n |x_n|$.

Of the norm axioms, only the triangle inequality for the p-norm is not immediate. The next two theorems supply it.

Theorem 2.3 (Hölder's inequality)**.** Let $1 < p < \infty$ and let q be its *conjugate exponent*, defined by $\frac{1}{p} + \frac{1}{q} = 1$. Then for all $x, y \in \mathbb{K}^n$,

$$\|x \cdot y\|_1 \;=\; \sum_{j=1}^{n} |x_j y_j| \;\le\; \Big(\sum_{j=1}^{n} |x_j|^p\Big)^{1/p} \Big(\sum_{j=1}^{n} |y_j|^q\Big)^{1/q} \;=\; \|x\|_p \|y\|_q, \tag{2.1}$$

where $x \cdot y = (x_1 y_1, \dots, x_n y_n)$. The inequality also holds for the endpoint pair $(p, q) = (1, \infty)$ and, by symmetry, for $(\infty, 1)$.

Remark 2.4. For $p = q = 2$ the inequality is the Cauchy–Schwarz inequality.

Proof. The endpoint case is immediate: $\sum_j |x_j y_j| \le (\max_j |y_j|) \sum_j |x_j| = \|x\|_1 \|y\|_\infty$.

Let now $1 < p < \infty$. The first step is Young's inequality: for all $a, b \ge 0$,

$$ab \le \frac{a^p}{p} + \frac{b^q}{q}. \tag{2.2}$$

If $a = 0$ or $b = 0$ the left side vanishes and the right side is non-negative, so assume $a, b > 0$ and put $u_1 = a^p > 0$, $u_2 = b^q > 0$, $\alpha_1 = \frac{1}{p}$, $\alpha_2 = \frac{1}{q}$, so that $\alpha_1 + \alpha_2 = 1$. The logarithm is concave on $(0, \infty)$, since $(\log)''(t) = -1/t^2 < 0$, and concavity applied to the convex combination $\alpha_1 u_1 + \alpha_2 u_2$ gives

$$\log\big(u_1^{\alpha_1} u_2^{\alpha_2}\big) = \alpha_1 \log u_1 + \alpha_2 \log u_2 \le \log(\alpha_1 u_1 + \alpha_2 u_2).$$

The logarithm is increasing, so $u_1^{\alpha_1} u_2^{\alpha_2} \le \alpha_1 u_1 + \alpha_2 u_2$, which is exactly (2.2), because $u_1^{\alpha_1} u_2^{\alpha_2} = (a^p)^{1/p} (b^q)^{1/q} = ab$.

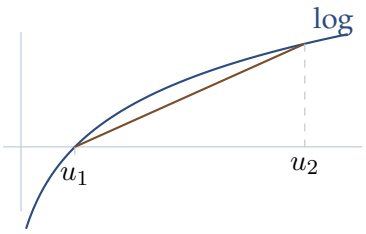


Figure 2.1. *Concavity of the logarithm: the chord over u_1 and u_2 lies below the graph, so $\alpha_1 \log u_1 + \alpha_2 \log u_2 \le \log(\alpha_1 u_1 + \alpha_2 u_2)$.*

The second step is a reduction by scaling. If $x = 0$ or $y = 0$ both sides of (2.1) vanish, so assume $\|x\|_p > 0$ and $\|y\|_q > 0$ and put $\tilde{x} = x/\|x\|_p$ and $\tilde{y} = y/\|y\|_q$, so that $\|\tilde{x}\|_p = \|\tilde{y}\|_q = 1$. Both sides of (2.1) are positively homogeneous of degree one in x and in y separately, so (2.1) for the pair x, y is equivalent to (2.1) for the normalised pair $\tilde{x}, \tilde{y}$. It therefore suffices to treat the case $\sum_j |x_j|^p = \sum_j |y_j|^q = 1$.

In that case, applying (2.2) with $a = |x_j|$ and $b = |y_j|$ and summing over j,

$$\sum_{j=1}^{n} |x_j y_j| \le \sum_{j=1}^{n} \Big(\frac{|x_j|^p}{p} + \frac{|y_j|^q}{q} \Big) = \frac{1}{p} + \frac{1}{q} = 1 = \|x\|_p \|y\|_q. \qquad \blacksquare$$

Theorem 2.5 (Minkowski's inequality)**.** Let $1 \le p < \infty$. Then $\|x + y\|_p \le \|x\|_p + \|y\|_p$ for all $x, y \in \mathbb{K}^n$.

Proof. For $p = 1$ the claim is the triangle inequality in $\mathbb{K}$ applied termwise. Let $1 < p < \infty$ and let q be the conjugate exponent, so that $q = \frac{p}{p-1}$ and hence $(p-1)q = p$. If $\sum_j |x_j + y_j|^p = 0$ the left side vanishes and the claim is trivial, so assume the sum is positive; being a finite sum, it is also finite. Using $|x_j + y_j| \le |x_j| + |y_j|$ and then Hölder (2.1) on each of the two resulting sums, with exponents q and p,

$$\begin{aligned}\sum_j |x_j + y_j|^p &\le \sum_j |x_j + y_j|^{p-1}|x_j| + \sum_j |x_j + y_j|^{p-1}|y_j| \\ &\le \Big(\sum_j |x_j + y_j|^{(p-1)q}\Big)^{1/q} \Big[\Big(\sum_j |x_j|^p\Big)^{1/p} + \Big(\sum_j |y_j|^p\Big)^{1/p}\Big] \\ &= \Big(\sum_j |x_j + y_j|^p\Big)^{1/q} \big(\|x\|_p + \|y\|_p\big).\end{aligned}$$

Dividing by the positive finite number $(\sum_j |x_j + y_j|^p)^{1/q}$ and using $1 - \frac{1}{q} = \frac{1}{p}$,

$$\|x + y\|_p = \Big(\sum_j |x_j + y_j|^p\Big)^{1/p} \le \|x\|_p + \|y\|_p. \qquad \blacksquare$$

2.1. Infinite sequences and functions

Both inequalities carry over from $\mathbb{K}^n$ to ℓ_p and to $C[0,1]$.

Theorem 2.6. Let $1 < p < \infty$ and $\frac{1}{p} + \frac{1}{q} = 1$. For all $x \in \ell_p$ and $y \in \ell_q$, $\|x \cdot y\|_1 \le \|x\|_p \|y\|_q$, and for all $f, g \in C[0,1]$,

$$\|fg\|_1 = \int_0^1 |fg| \le \Big(\int_0^1 |f|^p\Big)^{1/p} \Big(\int_0^1 |g|^q\Big)^{1/q} = \|f\|_p \|g\|_q.$$

Theorem 2.7. Let $1 \le p < \infty$. For all $x, y \in \ell_p$, $\|x+y\|_p \le \|x\|_p + \|y\|_p$, and for all $f, g \in C[0,1]$, $\|f + g\|_p \le \|f\|_p + \|g\|_p$.

2.2. The spaces $L_p[0,1]$

Definition 2.8 ($L_p[0,1]$)**.** Let $1 \le p < \infty$. On $C[0,1]$ the p-norm is a norm, but the space $(C[0,1], \|\cdot\|_p)$ is not complete. Its completion, in the sense of Proposition 1.5, is denoted $L_p[0,1]$.

Remark 2.9. The space $L_p[0,1]$ has a concrete description. Its elements are represented by those measurable functions $f\colon [0,1] \to \mathbb{K}$ for which $\int_0^1 |f|^p < \infty$, and two such functions represent the same element exactly when they agree outside a set of measure zero.

Remark 2.10. More generally, let $(X, \mathcal{A}, \mu)$ be a measure space and $1 \le p < \infty$. Then

$$L_p(X, \mathcal{A}, \mu) = \Big\{ f\colon X \to \mathbb{K} \,\Big|\, f \text{ measurable}, \int_X |f|^p \mathrm{d}\mu < \infty \Big\}, \qquad \|f\|_p = \Big(\int_X |f|^p \mathrm{d}\mu\Big)^{1/p},$$

and $L_p(X, \mathcal{A}, \mu)$ is a Banach space.

Exercise 2.11. Let $1 \le p < p' < \infty$. Show that $\ell_p \subset \ell_{p'}$.

Solution. Let $x \in \ell_p$. The series $\sum_j |x_j|^p$ converges, so $|x_j| \to 0$, and there is an N with $|x_j| < 1$ for all $j > N$. For such j the larger exponent gives the smaller value, $|x_j|^{p'} \le |x_j|^p$, whence

$$\sum_{j>N} |x_j|^{p'} \le \sum_{j>N} |x_j|^p < \infty.$$

The remaining finitely many terms are finite, so $\sum_j |x_j|^{p'} < \infty$, that is $x \in \ell_{p'}$. In particular $\ell_1 \subset \ell_{3/2} \subset \ell_2 \subset \cdots \subset \ell_\infty$. ◊

Exercise 2.12. Let $1 \le p < p' < \infty$. Show that $L_p[0,1] \supset L_{p'}[0,1]$, the inclusion running the other way than in Exercise 2.11.

Solution. Work with the representatives of Remark 2.9, so that $f \in L_{p'}[0,1]$ means $\int_0^1 |f|^{p'} < \infty$. Split the domain according to the size of $|f|$:

$$\int_0^1 |f|^p = \int_{\{|f|>1\}} |f|^p + \int_{\{|f|\le 1\}} |f|^p \le \int_0^1 |f|^{p'} + 1 < \infty.$$

On the first set $|f|^p \le |f|^{p'}$, since $p < p'$ and $|f| > 1$ there; the second integral is at most the measure of $[0,1]$, which is 1. Hence $f \in L_p[0,1]$. ◊

3. Equivalent norms

Definition 3.1 (Equivalent norms)**.** Two norms $\|\cdot\|_a$ and $\|\cdot\|_b$ on a vector space X are *equivalent* if there are constants $c, C > 0$ such that $c\|x\|_a \le \|x\|_b \le C\|x\|_a$ for all $x \in X$.

Equivalent norms induce the same topology, hence the same convergent sequences with the same limits, and the same Cauchy sequences. As a rule of thumb, the norms occurring naturally on an infinite-dimensional space are not equivalent; the finite-dimensional case is the opposite extreme.

Lemma 3.2. Let $(X, \|\cdot\|)$ be a normed space over $\mathbb{K}$ and let $x_1, \dots, x_n \in X$ be linearly independent. Then there is a $C > 0$ such that

$$\Big\|\sum_{i=1}^n \alpha_i x_i\Big\| \;\ge\; C \sum_{i=1}^n |\alpha_i| \qquad \text{for all } \alpha_1, \dots, \alpha_n \in \mathbb{K}.$$

Proof. Both sides are positively homogeneous of degree one in $(\alpha_1, \dots, \alpha_n)$, and both vanish when every α_i is zero, so it suffices to find a $C > 0$ that works for all coefficient vectors with $\sum_i |\alpha_i| = 1$.

Suppose no such C exists. Then for every $m \in \mathbb{N}$ there are coefficients $\alpha_1^{(m)}, \dots, \alpha_n^{(m)}$ with $\sum_i |\alpha_i^{(m)}| = 1$ and

$$\Big\|\sum_{i=1}^n \alpha_i^{(m)} x_i\Big\| < \frac{1}{m}.$$

Each coordinate sequence $(\alpha_i^{(m)})_m$ is bounded by 1, so by the Bolzano–Weierstrass theorem it has a convergent subsequence. Extracting successively for $i = 1, \dots, n$ and passing to the last subsequence, we may assume $\alpha_i^{(m)} \to \alpha_i$ for every i simultaneously. Then $\sum_i |\alpha_i| = \lim_m \sum_i |\alpha_i^{(m)}| = 1$, so not every α_i is zero. The map $(\alpha_1, \dots, \alpha_n) \mapsto \sum_i \alpha_i x_i$ is continuous, and so is the norm, hence

$$\Big\| \sum_{i=1}^{n} \alpha_i x_i \Big\| = \lim_{m \to \infty} \Big\| \sum_{i=1}^{n} \alpha_i^{(m)} x_i \Big\| = 0,$$

so $\sum_i \alpha_i x_i = 0$ with coefficients not all zero. This contradicts the independence of $x_1, \dots, x_n$. ■

Theorem 3.3. If X is a finite-dimensional vector space over $\mathbb{K}$, then any two norms on X are equivalent.

Proof. Let $x_1, \dots, x_n$ be a basis of X and, for $x = \sum_{i=1}^n \alpha_i x_i$, set $\|x\|_1 = \sum_{i=1}^n |\alpha_i|$, which is a norm on X. Let $\|\cdot\|$ be any norm on X. By Lemma 3.2 there is a $C > 0$ with $\|x\| \geq C\|x\|_1$ for all $x \in X$. In the other direction,

$$\|x\| = \Big\| \sum_{i=1}^{n} \alpha_i x_i \Big\| \leq \sum_{i=1}^{n} |\alpha_i| \|x_i\| \leq \Big(\max_{1 \leq i \leq n} \|x_i\| \Big) \sum_{i=1}^{n} |\alpha_i| = c' \|x\|_1,$$

with $c' = \max_{1 \leq i \leq n} \|x_i\| > 0$, the maximum being positive because the basis vectors are non-zero. So every norm on X is equivalent to $\|\cdot\|_1$.

Equivalence of norms is transitive: if $c_a \|x\|_1 \leq \|x\|_a \leq C_a \|x\|_1$ and $c_b \|x\|_1 \leq \|x\|_b \leq C_b \|x\|_1$ for all x, then

$$\frac{c_b}{C_a} \|x\|_a \leq \|x\|_b \leq \frac{C_b}{c_a} \|x\|_a.$$

Applying this to two arbitrary norms on X, each equivalent to $\|\cdot\|_1$ by the first part, gives the claim. ■

4. Separable spaces

Definition 4.1 (Countable set)**.** A set S is *countable* if it is finite or can be listed as $S = \{s_1, s_2, s_3, \dots\}$.

Definition 4.2 (Dense set)**.** A set $S \subset (X, d)$ is *dense* if $\overline{S} = X$.

Definition 4.3 (Separable space)**.** A metric space (X, d) is *separable* if it contains a countable dense set.

Remark 4.4. $S \subset (X, d)$ is dense if and only if for every $x \in X$ there is a sequence $(s_n) \subset S$ with $s_n \to x$.

The spaces introduced so far are now checked for completeness and for separability.

Example 4.5. $(\mathbb{R}, |\cdot|)$ is Banach and separable, with $\mathbb{Q}$ as a countable dense set.

Example 4.6. Let $\mathbb{K} = \mathbb{R}$ or $\mathbb{C}$. Then $(\mathbb{K}^n, \|\cdot\|_2)$ is Banach and separable: $\mathbb{Q}^n$ is a countable dense set for $\mathbb{K} = \mathbb{R}$, and $(\mathbb{Q} + i\mathbb{Q})^n$ for $\mathbb{K} = \mathbb{C}$.

Example 4.7. $(\mathbb{K}^n, \|\cdot\|_p)$ for $1 \le p < \infty$ and $(\mathbb{K}^n, \|\cdot\|_\infty)$ are Banach and separable, because these norms are equivalent to $\|\cdot\|_2$ by Theorem 3.3, and equivalent norms have the same convergent sequences and the same dense sets.

Remark 4.8. On a finite-dimensional space any such argument may be carried out in the Euclidean norm, since all norms there are equivalent.

Example 4.9. $(\ell_p, \|\cdot\|_p)$ is separable for $1 \le p < \infty$. The countable dense set consists of the finitely supported rational sequences $q = (q_1, \dots, q_m, 0, 0, \dots)$ with $q_i \in \mathbb{Q}$, or with $q_i \in \mathbb{Q} + i\mathbb{Q}$ when $\mathbb{K} = \mathbb{C}$: an $x = (x_1, x_2, \dots) \in \ell_p$ is approximated by truncating it and then approximating the first m coordinates.

Theorem 4.10. $(\ell_p, \|\cdot\|_p)$ is a Banach space for $1 \le p < \infty$.

Proof. Let $(x_m) \subset \ell_p$ be Cauchy, $x_m = (\xi_{1,m}, \xi_{2,m}, \dots)$. Given $\varepsilon > 0$ there is an N such that

$$\Big(\sum_{j=1}^{\infty} |\xi_{j,m} - \xi_{j,n}|^p\Big)^{1/p} \le \varepsilon \qquad \text{for all } n, m > N.$$

A single term is at most the whole sum, so $|\xi_{j,m} - \xi_{j,n}| \le \varepsilon$ for every j and all $n, m > N$. Thus each coordinate sequence $(\xi_{j,m})_m$ is Cauchy in $\mathbb{K}$ and converges; write $\xi_j = \lim_{m\to\infty} \xi_{j,m}$ and $x = (\xi_1, \xi_2, \dots)$.

Fix k. For all $n, m > N$ the partial sum satisfies $\sum_{j=1}^{k} |\xi_{j,m} - \xi_{j,n}|^p \le \varepsilon^p$. Fix $m > N$ and let $n \to \infty$; the sum is finite, so the limit may be taken termwise, and

$$\sum_{j=1}^{k} |\xi_{j,m} - \xi_j|^p \le \varepsilon^p.$$

The bound does not depend on k, so letting $k \to \infty$ gives $\sum_{j=1}^{\infty} |\xi_{j,m} - \xi_j|^p \le \varepsilon^p$, that is $x_m - x \in \ell_p$ and $\|x_m - x\|_p \le \varepsilon$ for every $m > N$.

Consequently $x = (x - x_m) + x_m$ lies in ℓ_p, by Theorem 2.7 and $x_m \in \ell_p$. The estimate $\|x_m - x\|_p \le \varepsilon$ for all $m > N$ is convergence in the norm, not merely coordinatewise, so $x_m \to x$ in ℓ_p. ∎

Counterexample 4.11. $(C[0,1], \|\cdot\|_p)$ is not complete, for $1 \le p < \infty$.

Discussion. A first attempt takes $f_n(x) = x^n$. These are continuous, and

$$\|f_n - 0\|_p = \Big(\int_0^1 x^{np} \mathrm{d}x\Big)^{1/p} = \Big(\frac{1}{np+1}\Big)^{1/p} \longrightarrow 0,$$

so (f_n) is Cauchy but converges in $\|\cdot\|_p$ to the function 0, which is continuous. The attempt establishes nothing: a Cauchy sequence whose limit is continuous is no obstruction to completeness.

Take instead the ramps

$$f_n(x) = \begin{cases} 0, & 0 \le x \le \frac{1}{2} - \frac{1}{n}, \\ n\big(x - \frac{1}{2} + \frac{1}{n}\big), & \frac{1}{2} - \frac{1}{n} \le x \le \frac{1}{2}, \\ 1, & \frac{1}{2} \le x \le 1, \end{cases} \qquad n \ge 3.$$

For $n < m$ the functions f_n and f_m agree outside $[\frac{1}{2} - \frac{1}{n}, \frac{1}{2}]$ and differ by at most 1 on it, so $\|f_n - f_m\|_p \le n^{-1/p}$ and (f_n) is Cauchy in $\|\cdot\|_p$.

Suppose $g \in C[0,1]$ satisfied $\|f_n - g\|_p \to 0$. Fix $\delta > 0$. For $n > 1/\delta$ the function f_n vanishes on $[0, \frac{1}{2} - \delta]$, so

$$\int_0^{1/2-\delta} |g|^p \le \|f_n - g\|_p^p \longrightarrow 0,$$

whence $g \equiv 0$ on $[0, \frac{1}{2} - \delta]$, and since $\delta > 0$ was arbitrary, $g \equiv 0$ on $[0, \frac{1}{2})$; continuity gives $g(\frac{1}{2}) = 0$. The same estimate on $[\frac{1}{2}, 1]$, where $f_n \equiv 1$, gives $g \equiv 1$ there, so $g(\frac{1}{2}) = 1$. This is a contradiction, so no continuous limit exists. ◊

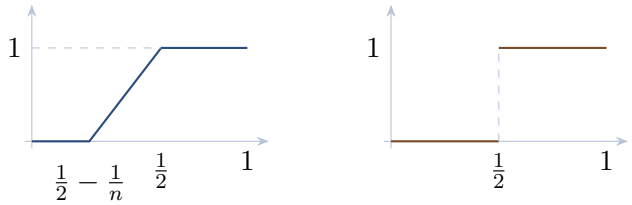


Figure 4.1. *The ramps f_n and the step function they approach. The step function is not continuous, so the limit leaves $C[0,1]$.*

Example 4.12. $L_p[0,1]$ and $L_p(\mathbb{R})$ are Banach and separable for $1 \le p < \infty$. For $L_p[0,1]$ either description may be used: the completion of $(C[0,1], \|\cdot\|_p)$, or the Lebesgue measurable f with $(\int |f|^p)^{1/p} < \infty$. Separability comes from the polynomials with rational coefficients, $p(x) = q_m x^m + \cdots + q_1 x + q_0$ with $q_i \in \mathbb{Q}$, which are dense by Bernstein's theorem.

Theorem 4.13. $(C[0,1], \|\cdot\|_\infty)$ is a Banach space.

Proof. Let $(f_n) \subset C[0,1]$ be a Cauchy sequence. Given $\varepsilon > 0$ there is an N such that $\|f_n - f_m\|_\infty = \max_{x\in[0,1]} |f_n(x) - f_m(x)| < \varepsilon$ for all $n, m > N$, and hence $|f_n(x) - f_m(x)| < \varepsilon$ for every x. So for fixed $x \in [0,1]$ the sequence $(f_n(x))$ is Cauchy in $\mathbb{K}$ and converges; write $f(x) = \lim_{n\to\infty} f_n(x)$. It remains to show that f is continuous and that $f_n \to f$ uniformly.

Let $n > N$. For all $m > N$ one has $|f_n(x) - f_m(x)| < \varepsilon$; letting $m \to \infty$ gives $|f_n(x) - f(x)| \le \varepsilon$ for all $n > N$ and all $x \in [0,1]$, which is uniform convergence.

Now fix $x \in [0,1]$ and $\varepsilon > 0$, and choose N with $|f_N(z) - f(z)| < \frac{\varepsilon}{3}$ for all $z \in [0,1]$. The function f_N is continuous on the compact interval $[0,1]$, hence uniformly continuous, so there is a $\delta > 0$ with $|f_N(x) - f_N(y)| < \frac{\varepsilon}{3}$ whenever $|x - y| < \delta$. For such x, y,

$$|f(x) - f(y)| \le |f(x) - f_N(x)| + |f_N(x) - f_N(y)| + |f_N(y) - f(y)| < \frac{\varepsilon}{3} + \frac{\varepsilon}{3} + \frac{\varepsilon}{3} = \varepsilon,$$

so $f \in C[0,1]$. ■

Example 4.14. $(C[0,1], \|\cdot\|_\infty)$ is also separable: the separability argument of Example 4.12, with polynomials with rational coefficients, applies here as well.

Example 4.15. Let $C_p^r[0,1]$ carry $\|f\| := \|f\|_p + \|f'\|_p + \cdots + \|f^{(r)}\|_p$. This is a norm, but the space is not complete. Its completion is the Sobolev space $L_p^r[0,1]$.

5. Compact sets

Definition 5.1 (Compactness)**.** A set $K \subset (X,d)$ is *compact* if every open covering of K admits a finite subcovering.

Theorem 5.2. For $K \subset (X,d)$ the following are equivalent.

1. K is compact.
2. Every sequence in K has a convergent subsequence whose limit lies in K.
3. K is complete and totally bounded, that is, for every $\varepsilon > 0$ there are finitely many balls $B(x_1,\varepsilon),\dots,B(x_N,\varepsilon)$ with $K \subset \bigcup_{i=1}^{N} B(x_i,\varepsilon)$.

Lemma 5.3. A compact $K \subset (X,\|\cdot\|)$ is closed and bounded.

Proof. For closedness, let $(x_n) \subset K$ with $x_n \to x$. By Theorem 5.2 the sequence has a subsequence converging to a point of K, and that limit is x, so $x \in K$.

For boundedness, suppose K were unbounded. Then there is a sequence $(x_n) \subset K$ with $\|x_n\| \geq n$, which has no convergent subsequence, contradicting Theorem 5.2. ∎

Theorem 5.4 (Heine–Borel)**.** A set $K \subset \mathbb{R}^n$ is compact if and only if it is closed and bounded.

Definition 5.5 (Local compactness)**.** A metric space (X,d) is *locally compact* if every $x \in X$ has a neighbourhood $B(x,\varepsilon)$ whose closure $\overline{B(x,\varepsilon)}$ is compact.

Definition 5.6 (Relative compactness)**.** A set $S \subset (X,d)$ is *relatively compact* if $\overline{S}$ is compact.

Lemma 5.7 (Riesz)**.** Let $M \subset (X,\|\cdot\|)$ be a proper closed subspace and let $t \in (0,1)$. Then there is an $x_t \in X$ with $\|x_t\| = 1$ and $\operatorname{dist}(x_t, M) \geq t$, where $\operatorname{dist}(x,M) = \inf\{\|x-m\| : m \in M\}$.

Proof. Since M is a proper subspace there is a $u \in X \setminus M$, and since M is closed, $d := \operatorname{dist}(u,M) > 0$. As $t \in (0,1)$ we have $d/t > d$, so by the definition of the infimum there is an $x_0 \in M$ with

$$d \leq \|u - x_0\| \leq \frac{d}{t}.$$

Put $x_t = (u-x_0)/\|u-x_0\|$, a unit vector. Let $x \in M$ be arbitrary and set $x_1 = \|u-x_0\|\,x + x_0$, which lies in M because M is a subspace. Then

$$\|x - x_t\| = \frac{1}{\|u-x_0\|}\big\|\|u-x_0\|\,x - u + x_0\big\| = \frac{\|x_1 - u\|}{\|u-x_0\|} \geq \frac{d}{\|u-x_0\|} \geq \frac{d}{d/t} = t,$$

using $\|x_1 - u\| \geq \operatorname{dist}(u,M) = d$. As $x \in M$ was arbitrary, $\operatorname{dist}(x_t, M) \geq t$. ∎

Theorem 5.8. A normed space $(X, \|\cdot\|)$ is locally compact if and only if $\dim X < \infty$.

Proof. For $x \in X$ and $\varepsilon > 0$ the map $y \mapsto x + \varepsilon y$ is a homeomorphism of X carrying $\overline{B(0,1)}$ onto $\overline{B(x,\varepsilon)}$. All closed balls are therefore compact together with $\overline{B(0,1)}$, and X is locally compact if and only if $\overline{B(0,1)}$ is compact.

Assume $\dim X = \infty$. Choose a unit vector $x_1 \in X$ and put $M_1 = \operatorname{span}\{x_1\}$. Each $M_i = \operatorname{span}\{x_1, \dots, x_i\}$ is finite-dimensional, hence closed: by Theorem 3.3 its norm is equivalent to the norm transported from $\mathbb{K}^i$, which is complete, so M_i is complete and therefore closed in X. It is proper because $\dim X = \infty$. So Lemma 5.7 with $t = \frac{1}{2}$ yields a unit vector x_{i+1} with $\operatorname{dist}(x_{i+1}, M_i) \geq \frac{1}{2}$. This produces a sequence of unit vectors with

$$\|x_n - x_m\| \geq \tfrac{1}{2} \qquad \text{for } n \neq m,$$

since for $n > m$ one has $x_m \in M_{n-1}$. The sequence lies in $\overline{B(0,1)}$ and has no convergent subsequence, so $\overline{B(0,1)}$ is not compact by Theorem 5.2, and X is not locally compact.

Assume $\dim X = n < \infty$. Fixing a basis identifies X with $\mathbb{K}^n$, and by Theorem 3.3 the norm of X is equivalent to the Euclidean norm transported by that identification, so the two have the same closed sets and the same bounded sets. The set $\overline{B(0,1)}$ is closed and bounded, hence compact by Theorem 5.4, and X is locally compact. ∎

Corollary 5.9. $\overline{B(0,1)} \subset (X, \|\cdot\|)$ is compact if and only if $\dim X < \infty$.

6. Compact sets in $(C[a,b], \|\cdot\|_\infty)$

Definition 6.1 (Uniform boundedness)**.** A set $S \subset C[a,b]$ is *uniformly bounded* if there is a $K > 0$ with $\|f\|_\infty \leq K$ for every $f \in S$.

Definition 6.2 (Uniform equicontinuity)**.** A set $S \subset C[a,b]$ is *uniformly equicontinuous* if for every $\varepsilon > 0$ there is a $\delta > 0$ such that for every $f \in S$ and all $x, y \in [a,b]$ with $|x - y| < \delta$ one has $|f(x) - f(y)| < \varepsilon$.

Theorem 6.3 (Arzelà–Ascoli)**.** A set $S \subset (C[a,b], \|\cdot\|_\infty)$ is relatively compact if and only if it is uniformly bounded and uniformly equicontinuous. If S is in addition closed, it is compact.

Proof. Assume first that S is relatively compact. Then $\overline{S}$ is compact, hence bounded by Lemma 5.3, so S is uniformly bounded. Fix $\varepsilon > 0$. By Theorem 5.2 the compact set $\overline{S}$ is totally bounded, so there are finitely many $f_1, \dots, f_r$ with $\overline{S} \subset \bigcup_{i=1}^{r} B(f_i, \frac{\varepsilon}{3})$. Each f_i is continuous on the compact interval $[a,b]$, hence uniformly continuous, so there is a $\delta_i > 0$ with $|f_i(x) - f_i(y)| < \frac{\varepsilon}{3}$ whenever $|x - y| < \delta_i$. Put $\delta = \min_{1 \leq i \leq r} \delta_i$, which is positive because the minimum is over finitely many indices. Given $f \in S$, choose i with $\|f - f_i\|_\infty < \frac{\varepsilon}{3}$. Then for $|x - y| < \delta$,

$$|f(x) - f(y)| \leq |f(x) - f_i(x)| + |f_i(x) - f_i(y)| + |f_i(y) - f(y)| < \varepsilon,$$

so S is uniformly equicontinuous.

Assume now that S is uniformly bounded, by K, and uniformly equicontinuous, and let $(f_n) \subset S$. The goal is to select a convergent subsequence. Enumerate the rationals of the interval,

$\mathbb{Q} \cap [a,b] = \{x_1, x_2, \dots\}$. The values $f_n(x_1)$ satisfy $|f_n(x_1)| \le K$, so some subsequence $(f_n^{(1)})$ converges at x_1. Along it, the values at x_2 are again bounded, so a further subsequence $(f_n^{(2)})$ converges at x_2, and so on. Let $g_m = f_m^{(m)}$ be the diagonal sequence. It is a subsequence of the original (f_n), and from the index k onwards it is a subsequence of $(f_n^{(k)})$, so $(g_m(x_k))$ converges for every k.

The claim is that (g_m) is uniformly Cauchy. Fix $\varepsilon > 0$ and let $\delta > 0$ be given by uniform equicontinuity for $\frac{\varepsilon}{3}$. Cover $[a,b]$ by finitely many sets $U_1, \dots, U_s$ of diameter less than δ, and pick in each U_j a rational point x_{k_j}, which exists because the rationals are dense. For each j the sequence $(g_m(x_{k_j}))_m$ converges, hence is Cauchy, so there is an N_j with $|g_n(x_{k_j}) - g_m(x_{k_j})| < \frac{\varepsilon}{3}$ for $n, m > N_j$. Put $N = \max_{1 \le j \le s} N_j$, finite because there are finitely many j. Let $t \in [a,b]$ and choose j with $t \in U_j$; then $|t - x_{k_j}| < \delta$, so for $n, m > N$,

$$|g_n(t) - g_m(t)| \le |g_n(t) - g_n(x_{k_j})| + |g_n(x_{k_j}) - g_m(x_{k_j})| + |g_m(x_{k_j}) - g_m(t)| < \varepsilon.$$

Hence $\|g_n - g_m\|_\infty \le \varepsilon$ for $n, m > N$. The space $(C[a,b], \|\cdot\|_\infty)$ is complete, by the proof of Theorem 4.13 with $[a,b]$ in place of $[0,1]$, so (g_m) converges in it.

Finally, $\overline{S}$ has the same uniform bound K and the same modulus of equicontinuity as S, so the selection above applies verbatim to a sequence in $\overline{S}$, and its limit lies in $\overline{S}$ because $\overline{S}$ is closed. By Theorem 5.2 the set $\overline{S}$ is compact, that is, S is relatively compact. ∎

7. Bases

Definition 7.1 (Algebraic basis)**.** Let X be a vector space. A set $S \subset X$ is an *algebraic basis*, or *Hamel basis*, if S is linearly independent and every $x \in X$ has a representation $x = \sum_{i \in I} \alpha_i s_i$ with $s_i \in S$, $\alpha_i \in \mathbb{K}$ and I finite.

Example 7.2. Let $1 \le p < \infty$ and $e_n = (0, \dots, 0, 1, 0, \dots)$, with the 1 in position n. The set $E = \{e_1, e_2, \dots\}$ is not an algebraic basis of ℓ_p: a finite combination of the e_n has only finitely many non-zero coordinates, while $(2^{-n})_n \in \ell_p$ has infinitely many.

Definition 7.3 (Convergent series)**.** In a normed space X a series $\sum_{n=1}^{\infty} x_n$ *converges* to $x \in X$ if its partial sums $s_N := \sum_{n=1}^{N} x_n$ satisfy $s_N \to x$.

Definition 7.4 (Schauder basis)**.** In a Banach space X a sequence (s_n) is a *Schauder basis* if every $x \in X$ has a unique representation $x = \sum_{i=1}^{\infty} \alpha_i s_i$ with $\alpha_i \in \mathbb{K}$, the series converging in the sense of Definition 7.3.

Example 7.5. $E = (e_n)$ is a Schauder basis of ℓ_p for $1 \le p < \infty$.

Proof. Let $x = (\alpha_1, \alpha_2, \dots) \in \ell_p$ and $s_N = \sum_{i=1}^{N} \alpha_i e_i = (\alpha_1, \dots, \alpha_N, 0, \dots)$. Then

$$\|x - s_N\|_p^p = \|(0, \dots, 0, \alpha_{N+1}, \alpha_{N+2}, \dots)\|_p^p = \sum_{i=N+1}^{\infty} |\alpha_i|^p \longrightarrow 0,$$

being the tail of a convergent series, so $x = \sum_i \alpha_i e_i$. For uniqueness, suppose $x = \sum_i \beta_i e_i$. The j-th coordinate of the N-th partial sum is β_j once $N \ge j$, and a coordinate satisfies $|y_j| \le \|y\|_p$, so coordinates pass to the limit and $\beta_j = \alpha_j$. ∎

Remark 7.6. E is not a Schauder basis of ℓ_∞. Take $x = (1, 1, \dots)$. Since $|y_j| \le \|y\|_\infty$, the argument above shows that any representation $x = \sum_i \beta_i e_i$ must have $\beta_i = 1$ for all i. But then

$$\|x - s_N\|_\infty = \|(0, \dots, 0, 1, 1, \dots)\|_\infty = 1$$

for every N, so the partial sums do not converge to x, and x has no representation at all.

Remark 7.7. ℓ_∞ is not separable. The sequences with every term 0 or 1 form an uncountable set, and any two distinct ones have $\|\cdot\|_\infty$ distance 1. The balls of radius $\frac{1}{3}$ around them are therefore pairwise disjoint: a common point would put the two centres at distance at most $\frac{2}{3}$. A dense set meets every one of these balls, and meets distinct balls in distinct points, so it is uncountable.

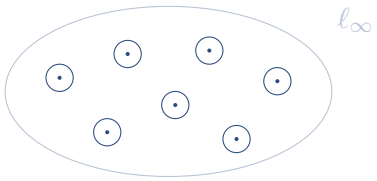


Figure 7.1. *Around the uncountably many* 0–1 *sequences, the balls of radius* $\frac{1}{3}$ *in* ℓ_∞ *are pairwise disjoint, and a dense set needs a point in each.*

Example 7.8. The Fourier systems are Schauder bases. In the real space $L_2[0, 2\pi]$ the system $1, \cos x, \sin x, \cos 2x, \sin 2x, \dots$ is one: every f has a unique representation

$$f = a_0 + \sum_{k=1}^{\infty} \big(a_k \cos(kx) + b_k \sin(kx)\big).$$

In the complex space $L_2[0, 2\pi]$ the system e^{ikx}, $k \in \mathbb{Z}$, enumerated as $1, e^{ix}, e^{-ix}, e^{2ix}, e^{-2ix}, \dots$, is one: every f has a unique representation $f = \sum_{k=-\infty}^{\infty} c_k e^{ikx}$, meaning

$$\Big\| f - \sum_{k=-N}^{N} c_k e^{ikx} \Big\|_2^2 = \int_0^{2\pi} \Big| f(x) - \sum_{k=-N}^{N} c_k e^{ikx} \Big|^2 \mathrm{d}x \xrightarrow[N\to\infty]{} 0.$$

Convergence in $\|\cdot\|_2$ does not guarantee pointwise convergence.

Proposition 7.9. A Banach space with a Schauder basis is separable.

Proof. Let (s_i) be a Schauder basis and let D be the set of finite sums $\sum_{i=1}^{N} q_i s_i$ with $N \in \mathbb{N}$ and $q_i \in \mathbb{Q}$, or $q_i \in \mathbb{Q} + i\mathbb{Q}$ when $\mathbb{K} = \mathbb{C}$. For each N these sums form a countable set, so D is countable. Given $x = \sum_i \alpha_i s_i$ and $\varepsilon > 0$, choose N with $\|x - \sum_{i=1}^{N} \alpha_i s_i\| < \frac{\varepsilon}{2}$, and then rational q_i with $|\alpha_i - q_i| \, \|s_i\| < \frac{\varepsilon}{2N}$ for $i \le N$. The triangle inequality gives $\|x - \sum_{i=1}^{N} q_i s_i\| < \varepsilon$, so D is dense. ∎

Corollary 7.10. ℓ_∞ has no Schauder basis.

Proof. It is not separable, by Remark 7.7, so Proposition 7.9 excludes a Schauder basis. ∎

Lemma 7.11 (Zorn)**.** Let $(P, \leq)$ be a partially ordered set in which every chain, that is every totally ordered subset, has an upper bound in P. Then P has a maximal element, an $m \in P$ such that $m \leq p$ implies $p = m$.

Remark 7.12. The *axiom of choice* states that $\prod_{i \in I} A_i$ is non-empty whenever every A_i is non-empty, that is, that there is a family $(a_i)_{i \in I}$ with $a_i \in A_i$ for every i.

Theorem 7.13. Every vector space has an algebraic basis.

Remark 7.14. An algebraic basis has no connection to any norm on the space.

Proof of Theorem 7.13. Let P be the family of linearly independent subsets of the vector space X, and write $I \leq J$ if $I \subseteq J$. This is a partial order: it is reflexive, $I \leq I$; antisymmetric, $I \leq J$ and $J \leq I$ give $I = J$; and transitive, $I \leq J$ and $J \leq K$ give $I \leq K$. The family is non-empty, since $\emptyset \in P$.

Let $\mathcal{C} \subset P$ be a chain. Its union U is linearly independent: finitely many $u_1, \ldots, u_k \in U$ lie in members $C_1, \ldots, C_k$ of $\mathcal{C}$, which are totally ordered, so the largest of them contains all u_j, and these are therefore independent. Thus $U \in P$ is an upper bound of $\mathcal{C}$; for the empty chain, $\emptyset$ is one. By Lemma 7.11 there is a maximal independent set S.

S spans X. Otherwise there is an $x \notin \operatorname{span} S$, and $S \cup \{x\}$ is independent: in a vanishing combination the coefficient of x must be 0, since otherwise $x \in \operatorname{span} S$, and then the remaining coefficients vanish by independence of S. This contradicts maximality, so S is an algebraic basis. ∎

Definition 7.15 (Absolute convergence)**.** A series $\sum_{i=1}^{\infty} x_i$ in a normed space X is *absolutely convergent* if $\sum_{i=1}^{\infty} \|x_i\| < \infty$.

Lemma 7.16 (Absolute convergence)**.** If X is a Banach space and $\sum_{i=1}^{\infty} x_i$ is absolutely convergent, then $\sum_{i=1}^{\infty} x_i$ converges.

Proof. As X is complete, it suffices to show that the partial sums $s_N = \sum_{i=1}^{N} x_i$ form a Cauchy sequence. Let $\varepsilon > 0$. The real series $\sum_i \|x_i\|$ converges, so there is an N with $\sum_{i=m+1}^{n} \|x_i\| < \varepsilon$ for all $n > m > N$. For such n, m,

$$\|s_n - s_m\| = \Big\| \sum_{i=m+1}^{n} x_i \Big\| \leq \sum_{i=m+1}^{n} \|x_i\| < \varepsilon. \qquad ∎$$

8. Bounded linear operators

Definition 8.1 (Linear operator)**.** Let X and Y be vector spaces over $\mathbb{K}$. A map $T \colon X \to Y$ is *linear* if $T(x_1 + x_2) = Tx_1 + Tx_2$ for all $x_1, x_2 \in X$ and $T(\alpha x) = \alpha\, Tx$ for all $\alpha \in \mathbb{K}$ and $x \in X$.

Definition 8.2 (Bounded operator, operator norm)**.** A linear operator $T\colon (X, \|\cdot\|) \to (Y, \|\cdot\|)$ between normed spaces is *bounded* if $\sup_{\|x\|\le 1}\|Tx\| < \infty$. In that case

$$\|T\| = \sup_{\|x\|\le 1} \|Tx\|$$

is the *operator norm* of T.

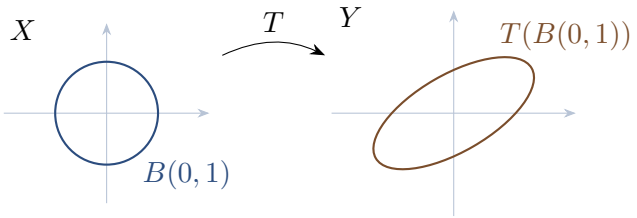


Figure 8.1. *A bounded operator maps the closed unit ball to a bounded set, here an ellipse; $\|T\|$ is the radius of the smallest ball about 0 containing the image.*

Remark 8.3. For every $x \in X$, $\|Tx\| \le \|T\|\,\|x\|$.

Proof. For $x = 0$ both sides vanish, since $T0 = 0$ by linearity. For $x \neq 0$, linearity gives $Tx = \|x\|\,T\big(x/\|x\|\big)$, and $x/\|x\|$ has norm 1, so

$$\|Tx\| = \|x\|\,\Big\|T\Big(\frac{x}{\|x\|}\Big)\Big\| \le \|x\| \sup_{\|x'\|\le 1}\|Tx'\| = \|x\|\,\|T\|. \qquad \blacksquare$$

Example 8.4. Every linear operator $T\colon X \to Y$ on a finite-dimensional normed space X is bounded. In particular every linear $T\colon \mathbb{K}^n \to \mathbb{K}^m$, which is given by an $m \times n$ matrix, is bounded.

Proof. Let $e_1, \dots, e_n$ be a basis of X. By Lemma 3.2 there is a $C > 0$ with $\|\sum_i \alpha_i e_i\| \ge C \sum_i |\alpha_i|$ for all coefficients. For $x = \sum_{i=1}^n \alpha_i e_i$,

$$\|Tx\| = \Big\|\sum_{i=1}^n \alpha_i\, Te_i\Big\| \le \sum_{i=1}^n |\alpha_i|\,\|Te_i\| \le \Big(\max_{1\le i\le n}\|Te_i\|\Big)\sum_{i=1}^n |\alpha_i| \le \frac{1}{C}\max_{1\le i\le n}\|Te_i\|\;\|x\|,$$

so $\|T\| \le C^{-1}\max_i\|Te_i\| < \infty$. $\blacksquare$

Example 8.5. For fixed $x \in \mathbb{K}^n$ let $f_x\colon \mathbb{K}^n \to \mathbb{K}$, $f_x(y) = \sum_{i=1}^n x_i y_i$. Then f_x is linear, and bounded by Example 8.4 with $m = 1$.

Definition 8.6 (Linear functional)**.** A linear operator from X to $\mathbb{K}$ is a *linear functional*.

Example 8.7 (Multiplication operator on $C[0,1]$)**.** Fix $f \in C[0,1]$ and let $M_f\colon C[0,1] \to C[0,1]$, $g \mapsto fg$, with the norm $\|\cdot\|_\infty$. Then M_f is linear and bounded, with $\|M_f\| \le \|f\|_\infty$.

Proof. $\|f\|_\infty < \infty$ because f is continuous on the compact $[0,1]$. For $\|g\|_\infty \le 1$,

$$\|M_f g\|_\infty = \sup_{x\in[0,1]} |f(x)g(x)| \le \|f\|_\infty\|g\|_\infty \le \|f\|_\infty. \qquad \blacksquare$$

Example 8.8 (Multiplication operator on ℓ_p)**.** Let $1 \le p < \infty$, fix $x = (x_1, x_2, \dots) \in \ell_\infty$, and let $M_x \colon \ell_p \to \ell_p$, $y = (y_1, y_2, \dots) \mapsto (x_1 y_1, x_2 y_2, \dots)$. Then M_x is linear and bounded, with $\|M_x\| \le \|x\|_\infty$.

Proof. Linearity is immediate. Termwise $|x_i y_i|^p \le \|x\|_\infty^p |y_i|^p$, and summing gives $\|M_x y\|_p \le \|x\|_\infty \|y\|_p$. This shows first that $M_x y \in \ell_p$, so that M_x is well defined, and then that $\|M_x y\|_p \le \|x\|_\infty$ whenever $\|y\|_p \le 1$. ∎

Remark 8.9. Multiplication operators are the simplest operators. They are the counterpart of diagonal matrices: $D = \operatorname{diag}(\alpha_1, \dots, \alpha_n)$ acts by $y \mapsto Dy = (\alpha_1 y_1, \dots, \alpha_n y_n)$, which is multiplication by $(\alpha_1, \dots, \alpha_n)$, and diagonalizing a matrix in linear algebra reduces it to such an operator.

Example 8.10 (Integral functional)**.** Let $I \colon C[0,1] \to \mathbb{K}$, $I(f) = \int_0^1 f(t)\mathrm{d}t$, with the norm $\|\cdot\|_\infty$. Then I is linear and bounded, with $\|I\| \le 1$: for $\|f\|_\infty \le 1$,

$$|If| = \left| \int_0^1 f(t)\mathrm{d}t \right| \le \int_0^1 |f(t)|\mathrm{d}t \le \int_0^1 \mathrm{d}t = 1.$$

Example 8.11 (Evaluation functional)**.** Fix $t_0 \in [0,1]$ and let $E \colon C[0,1] \to \mathbb{K}$, $Ef = f(t_0)$, with the norm $\|\cdot\|_\infty$. Then E is linear and bounded, with $\|E\| \le 1$: for $\|f\|_\infty \le 1$, $|Ef| = |f(t_0)| \le \|f\|_\infty \le 1$.

Example 8.12 (Integral operator)**.** Fix a *kernel* $K \in C([0,1] \times [0,1])$ and let $T_K \colon C[0,1] \to C[0,1]$,

$$(T_K f)(x) = \int_0^1 K(x,y) f(y)\mathrm{d}y,$$

with the norm $\|\cdot\|_\infty$ on both sides. This is the continuous analogue of matrix multiplication, $(Kf)_j = \sum_{m=1}^n K_{jm} f_m$. Then T_K is linear and bounded, with $\|T_K\| \le \|K\|_\infty$, where $\|K\|_\infty = \max_{[0,1]^2} |K|$ is finite because K is continuous on a compact set.

Proof. $T_K f$ is continuous: K is uniformly continuous on the compact square, and $|(T_K f)(x) - (T_K f)(x')| \le \sup_y |K(x,y) - K(x',y)|\, \|f\|_\infty$. Linearity is immediate. For $\|f\|_\infty \le 1$,

$$\|T_K f\|_\infty = \sup_{x \in [0,1]} \left| \int_0^1 K(x,y) f(y)\mathrm{d}y \right| \le \sup_{x \in [0,1]} \int_0^1 |K(x,y)|\, |f(y)|\mathrm{d}y \le \|K\|_\infty \int_0^1 \mathrm{d}y = \|K\|_\infty. \quad ∎$$

Example 8.13 (Differentiation)**.** Let X be the space of polynomials on $[0,1]$ with the norm $\|\cdot\|_\infty$, a subspace of $C[0,1]$; it is not a Banach space. Let $D \colon X \to X$, $f \mapsto f'$. Then D is linear but not bounded: for $f_n(t) = t^n$ one has $\|f_n\|_\infty = 1$ for all n, while $\|Df_n\|_\infty = \|nt^{n-1}\|_\infty = n \to \infty$.

Remark 8.14. As a rule of thumb, integration is bounded and differentiation is not.

9. The space of bounded operators

Theorem 9.1. A linear operator $T\colon (X,\|\cdot\|) \to (Y,\|\cdot\|)$ is bounded if and only if it is continuous.

Proof. Assume T is bounded. If $T = 0$ it is continuous, so let $\|T\| > 0$. Given $\varepsilon > 0$ put $\delta = \varepsilon/\|T\|$. For all $x_1, x_2 \in X$ with $\|x_1 - x_2\| < \delta$, Remark 8.3 gives

$$\|Tx_1 - Tx_2\| = \|T(x_1 - x_2)\| \le \|T\|\,\|x_1 - x_2\| < \|T\|\frac{\varepsilon}{\|T\|} = \varepsilon.$$

Assume T is continuous; only continuity at 0 will be used. Since $T0 = 0$, the choice $\varepsilon = 1$ gives a $\delta > 0$ with $\|Tx\| < 1$ whenever $\|x\| < \delta$. For $\|z\| \le 1$ the vector $\frac{\delta}{2}z$ has norm less than δ, so

$$\|Tz\| = \frac{2}{\delta}\Big\|T\Big(\frac{\delta}{2}z\Big)\Big\| < \frac{2}{\delta}.$$

Hence T is bounded, with $\|T\| \le 2/\delta$. ∎

Corollary 9.2. A linear operator $T\colon (X,\|\cdot\|) \to (Y,\|\cdot\|)$ is continuous everywhere if and only if it is continuous at 0.

Proof. The second half of the proof of Theorem 9.1 used only continuity at 0 to obtain boundedness, and the first half turns boundedness into continuity everywhere. ∎

Definition 9.3. $\mathcal{B}(X,Y)$ denotes the set of all bounded linear operators from X to Y.

Theorem 9.4. $\mathcal{B}(X,Y)$ is a vector space under pointwise operations, and the operator norm is a norm on it. Therefore $(\mathcal{B}(X,Y),\|\cdot\|)$ is a normed space.

Proof. Pointwise sums and scalar multiples of linear maps are linear, and the zero operator is bounded. For $S, T \in \mathcal{B}(X,Y)$, $\alpha \in \mathbb{K}$ and $\|x\| \le 1$,

$$\|(S+T)x\| \le \|Sx\| + \|Tx\| \le \|S\| + \|T\|, \qquad \|(\alpha T)x\| = |\alpha|\,\|Tx\| \le |\alpha|\,\|T\|,$$

so $S+T$ and αT are bounded, and $\mathcal{B}(X,Y)$ is a subspace of the vector space of linear maps from X to Y.

For the norm axioms, $\|T\| \ge 0$ is clear. If $\|T\| = 0$, then Remark 8.3 gives $\|Tx\| \le \|T\|\,\|x\| = 0$ for every x, so $T = 0$. Homogeneity follows from $\|\alpha T\| = \sup_{\|x\|\le 1}|\alpha|\,\|Tx\| = |\alpha|\,\|T\|$. For the triangle inequality, the triangle inequality in Y gives

$$\|T+S\| = \sup_{\|x\|\le 1}\|(T+S)x\| \le \sup_{\|x\|\le 1}\big(\|Tx\| + \|Sx\|\big) \le \sup_{\|x\|\le 1}\|Tx\| + \sup_{\|x\|\le 1}\|Sx\| = \|T\| + \|S\|. \quad ∎$$

Theorem 9.5. If $(Y,\|\cdot\|)$ is a Banach space, then so is $(\mathcal{B}(X,Y),\|\cdot\|)$.

Corollary 9.6. $(\mathcal{B}(X,\mathbb{K}),\|\cdot\|)$ is a Banach space for every normed space X.

Notation 9.7. $X' = (\mathcal{B}(X, \mathbb{K}), \|\cdot\|)$ is the *dual space* of X.

Proof of Theorem 9.5. Let $(T_n) \subset \mathcal{B}(X, Y)$ be a Cauchy sequence: for every $\varepsilon > 0$ there is an N with $\|T_n - T_m\| \leq \varepsilon$ for all $n, m > N$. For each $x \in X$, Remark 8.3 gives

$$\|T_n x - T_m x\| = \|(T_n - T_m)x\| \leq \|T_n - T_m\| \, \|x\| \leq \varepsilon \|x\|,$$

and since ε is arbitrary, $(T_n x)$ is Cauchy in Y. As Y is complete, the pointwise limit $Tx := \lim_{n\to\infty} T_n x$ exists. It is linear, because $T(\alpha x + \beta y) = \lim_n (\alpha T_n x + \beta T_n y) = \alpha Tx + \beta Ty$.

It remains to show that T is bounded and that $T_n \to T$ in norm. Fix $n > N$ and let $m \to \infty$ in $\|T_n x - T_m x\| \leq \varepsilon \|x\|$; continuity of the norm gives

$$\|(T_n - T)x\| \leq \varepsilon \|x\| \qquad \text{for all } x \in X.$$

Thus $T_n - T$ is bounded with $\|T_n - T\| \leq \varepsilon$, so $T = T_n - (T_n - T) \in \mathcal{B}(X, Y)$ by Theorem 9.4, and $\|T_n - T\| \leq \varepsilon$ for all $n > N$ means $T_n \to T$. ∎

Proposition 9.8 (Composition)**.** If $T \in \mathcal{B}(Y, Z)$ and $S \in \mathcal{B}(X, Y)$, then $TS \in \mathcal{B}(X, Z)$ and $\|TS\| \leq \|T\| \, \|S\|$.

Proof. TS is linear as a composition of linear maps. By Remark 8.3,

$$\|TS\| = \sup_{\|x\|\leq 1} \|T(Sx)\| \leq \sup_{\|x\|\leq 1} \|T\| \, \|Sx\| = \|T\| \sup_{\|x\|\leq 1} \|Sx\| = \|T\| \, \|S\|. \qquad ∎$$

Remark 9.9. In particular, for a Banach space X the space $\mathcal{B}(X, X)$ is a Banach algebra: a Banach space, by Theorem 9.5, which is also an algebra under composition, with $\|TS\| \leq \|T\| \, \|S\|$.

Definition 9.10 (Invertible operator)**.** $T \in \mathcal{B}(X, X)$ is *invertible* if T is a bijection from X to X and T^{-1} is bounded.

10. Invertible operators

Notation 10.1. $\mathcal{B}(X) = \mathcal{B}(X, X)$.

Definition 10.2 (Invertible operator, general case)**.** Extending Definition 9.10, an operator $T \in \mathcal{B}(X, Y)$ is *invertible* if T is a bijection from X to Y and $T^{-1} \in \mathcal{B}(Y, X)$.

Three criteria for invertibility follow.

Proposition 10.3 (Finite dimensions)**.** For a linear map $T\colon \mathbb{K}^n \to \mathbb{K}^n$, that is $T \in \operatorname{End}(\mathbb{K}^n)$, the following are equivalent.

1. T is injective.
2. T is surjective.
3. $\det T \neq 0$.
4. T is invertible.

Proof. The equivalence of (i), (ii) and (iii) is a result of linear algebra. An invertible T is bijective, so (iv) implies (i). Conversely, if (i), and hence (ii), holds, then T is a bijection, and T^{-1} is linear as the inverse of a linear bijection. It is bounded automatically, because $\mathbb{K}^n$ is finite-dimensional, by Example 8.4. So T is invertible. ∎

Proposition 10.4 (Neumann series)**.** Let X be a Banach space and $T \in \mathcal{B}(X)$ with $\|T\| < 1$. Then $I - T$ is invertible, and

$$(I - T)^{-1} = \sum_{k=0}^{\infty} T^k.$$

Remark 10.5. Many calculations with bounded linear operators are similar to those with complex numbers. For example, if $z \in \mathbb{C}$ and $|z| < 1$, then $\frac{1}{1-z} = \sum_{k=0}^{\infty} z^k$.

Proof of Proposition 10.4. Let $S_N = \sum_{k=0}^{N} T^k$. For $N > M$,

$$\|S_N - S_M\| = \Big\| \sum_{k=M+1}^{N} T^k \Big\| \le \sum_{k=M+1}^{N} \|T^k\| \le \sum_{k=M+1}^{N} \|T\|^k,$$

which is a piece of the tail of the series $\sum_{k=0}^{\infty} \|T\|^k$. That series converges because $\|T\| < 1$, so given $\varepsilon > 0$ the right side is less than ε for all $N > M$ large enough. Thus (S_N) is Cauchy in $\mathcal{B}(X)$. As X is a Banach space, so is $\mathcal{B}(X)$, by Theorem 9.5, and S_N converges; let $S = \lim_{N\to\infty} S_N \in \mathcal{B}(X)$. Norm convergence gives pointwise convergence, since $\|S_N y - Sy\| \le \|S_N - S\| \, \|y\|$ for every $y \in X$.

The sum telescopes, and the two orders of multiplication give the same result, $S_N(I - T) = (I - T)S_N = I - T^{N+1}$. For $x \in X$, $\|T^{N+1}x\| \le \|T\|^{N+1}\|x\| \to 0$, so

$$S(I - T)x = \lim_{N\to\infty} S_N(I - T)x = \lim_{N\to\infty} (I - T^{N+1})x = x.$$

The operator $I - T$ is bounded, hence continuous by Theorem 9.1, and may therefore be taken inside the limit:

$$(I - T)Sx = (I - T) \lim_{N\to\infty} S_N x = \lim_{N\to\infty} (I - T)S_N x = \lim_{N\to\infty} (I - T^{N+1})x = x.$$

So $S(I - T) = (I - T)S = I$. Hence $I - T$ is a bijection with inverse $S \in \mathcal{B}(X)$. ∎

Proposition 10.6. Let $T \in \mathcal{B}(X, Y)$ be surjective, and suppose there is a $b > 0$ such that $\|Tx\| \ge b\|x\|$ for all $x \in X$, that is, T is bounded away from zero. Then T is invertible, and $\|T^{-1}\| \le \frac{1}{b}$.

Proof. T is injective, that is $\ker T = \{0\}$: if $Tx = 0$, then $b\|x\| \le \|Tx\| = 0$, so $x = 0$. Being also surjective, T is a bijection, so T^{-1} exists, and it is linear as the inverse of a linear bijection. It remains to show $T^{-1} \in \mathcal{B}(Y, X)$. For every $y \in Y$ there is an $x \in X$ with $Tx = y$, and then

$$\|T^{-1}y\| = \|x\| \le \frac{1}{b}\|Tx\| = \frac{1}{b}\|y\|,$$

so $\|T^{-1}\| \le \frac{1}{b}$. ∎

11. Dual spaces

The dual space $X' = \mathcal{B}(X, \mathbb{K})$ of a normed space X was introduced in Notation **??**. In some cases X' can be described explicitly.

Remark 11.1. An equation $(X, \|\cdot\|) = (Y, \|\cdot\|')$ between normed spaces means that the two can be identified with each other: there is a natural linear bijection $i\colon X \to Y$ with $\|i(x)\|' = \|x\|$ for all $x \in X$.

Theorem 11.2 (Finite dimensions)**.** $(\mathbb{R}^n, \|\cdot\|_2)' = (\mathbb{R}^n, \|\cdot\|_2)$.

Proof. The identification is given by the standard inner product $\langle x, \gamma\rangle = \sum_{k=1}^n x_k\gamma_k$. Let f be a linear functional on $(\mathbb{R}^n, \|\cdot\|_2)$ and $e_1, \dots, e_n$ the standard basis. For $x = \sum_{k=1}^n \alpha_k e_k$ linearity gives $f(x) = \sum_{k=1}^n \alpha_k f(e_k)$. Let $\gamma_k = f(e_k)$; then

$$f(x) = \sum_{k=1}^n \alpha_k\gamma_k = \langle \alpha, \gamma\rangle,$$

where $\alpha = x$, since the coordinates of x in the standard basis are its entries. So every linear functional is represented by an inner product with a vector $\gamma \in \mathbb{R}^n$. Conversely, each γ defines the functional $f_\gamma = \langle \cdot\,, \gamma\rangle$, which is bounded by Example 8.4. The map $\gamma \mapsto f_\gamma$ is linear, and injective because $f_\gamma(e_k) = \gamma_k$. It is therefore a linear bijection between $\mathbb{R}^n$ and $(\mathbb{R}^n, \|\cdot\|_2)'$.

It remains to check $\|f\| = \|\gamma\|_2$. By the Cauchy–Schwarz inequality,

$$\|f\| = \sup_{\|x\|_2\le 1} |f(x)| = \sup_{\|x\|_2\le 1} |\langle x, \gamma\rangle| \le \sup_{\|x\|_2\le 1} \|x\|_2\|\gamma\|_2 \le \|\gamma\|_2.$$

If $\gamma = 0$ both sides vanish. Otherwise Cauchy–Schwarz attains equality at $x = \gamma/\|\gamma\|_2$, which has $\|x\|_2 = 1$ and $\langle x, \gamma\rangle = \|\gamma\|_2$. So $\|f\| \ge \|\gamma\|_2$, and hence $\|f\| = \|\gamma\|_2$. ∎

Theorem 11.3 (ℓ_p–ℓ_q duality)**.** Let $1 < p < \infty$ and $\frac{1}{p} + \frac{1}{q} = 1$. Then $(\ell_p)' = \ell_q$. Moreover $(\ell_1)' = \ell_\infty$, but $(\ell_\infty)' \ne \ell_1$.

Proof of the first statement. Let $\beta = (\beta_1, \beta_2, \dots) \in \ell_q$, and define f_β on ℓ_p by

$$x = (\alpha_1, \alpha_2, \dots) \longmapsto f_\beta(x) = \alpha_1\beta_1 + \alpha_2\beta_2 + \cdots = \sum_{k=1}^\infty \alpha_k\beta_k.$$

By Hölder's inequality, Theorem 2.6, $\sum_k |\alpha_k\beta_k| \le \|x\|_p\|\beta\|_q < \infty$, so the series converges absolutely. Thus f_β is well defined, it is linear, and

$$\|f_\beta\| = \sup_{\|x\|_p\le 1} |f_\beta(x)| = \sup_{\|x\|_p\le 1} \Big|\sum_k \alpha_k\beta_k\Big| \le \sup_{\|x\|_p\le 1} \|x\|_p\|\beta\|_q \le \|\beta\|_q.$$

Equality can be attained in Hölder's inequality. If $\beta = 0$ there is nothing to show. Otherwise, for each k choose $\sigma_k \in \mathbb{K}$ with $|\sigma_k| = 1$ and $\sigma_k\beta_k = |\beta_k|$, and put

$$\alpha_k = \frac{\sigma_k|\beta_k|^{q-1}}{\|\beta\|_q^{q-1}}.$$

From $\frac{1}{p}+\frac{1}{q}=1$ one gets $(q-1)p=q$, so $|\alpha_k|^p=|\beta_k|^q/\|\beta\|_q^q$ and $\|x\|_p=1$, while

$$\sum_k \alpha_k\beta_k = \frac{\sum_k|\beta_k|^q}{\|\beta\|_q^{q-1}} = \|\beta\|_q.$$

Hence $\|f_\beta\|\geq\|\beta\|_q$, and $\|f_\beta\|=\|\beta\|_q$.

It remains to check whether every $f\in(\ell_p)'$ arises as $f=f_\beta$. Let $f\in(\ell_p)'$, let e_k be the standard basis vectors, and let $\beta_k=f(e_k)$. First, $\beta\in\ell_q$. Fix N, choose σ_k as above, and let $x^{(N)}$ have coordinates $\sigma_k|\beta_k|^{q-1}$ for $k\leq N$ and 0 for $k>N$. By linearity of f on this finite sum, and by $(q-1)p=q$,

$$\sum_{k=1}^{N}|\beta_k|^q = f\big(x^{(N)}\big)\leq\|f\|\,\|x^{(N)}\|_p = \|f\|\Big(\sum_{k=1}^{N}|\beta_k|^q\Big)^{1/p}.$$

If the left side is positive, dividing by its $\frac{1}{p}$-th power gives $(\sum_{k=1}^{N}|\beta_k|^q)^{1/q}\leq\|f\|$, and the bound holds trivially otherwise. As N is arbitrary, $\beta\in\ell_q$.

Second, $f=f_\beta$. Let $x=(\alpha_1,\alpha_2,\dots)\in\ell_p$. By Example 7.5, $x=\sum_k\alpha_k e_k$ with convergence in ℓ_p, and f is continuous by Theorem 9.1, so

$$f(x)=f\Big(\sum_k\alpha_k e_k\Big)=\lim_{N\to\infty}\sum_{k=1}^{N}\alpha_k f(e_k)=\sum_k\alpha_k\beta_k=f_\beta(x).$$

The map $\beta\mapsto f_\beta$ is linear and, by the norm identity, norm-preserving, hence injective. It is surjective by the last step. So it is a linear norm-preserving bijection from ℓ_q onto $(\ell_p)'$, and the two spaces are identified in the sense of Remark 11.1. ∎

Remark 11.4. The analogous duality holds for function spaces: for $1<p<\infty$ and $\frac{1}{p}+\frac{1}{q}=1$, $(L_p[0,1])'=L_q[0,1]$.

11.1. Reflexivity

Remark 11.5 (The natural map into the bidual). For $x\in X$ define $g_x\colon X'\to\mathbb{K}$ by $g_x(f)=f(x)$. This is linear, and $|g_x(f)|=|f(x)|\leq\|f\|\,\|x\|$ by Remark 8.3, so $g_x\in X''=(X')'$ with

$$\|g_x\|=\sup_{\|f\|\leq1}|g_x(f)|=\sup_{\|f\|\leq1}|f(x)|\leq\|x\|.$$

The *natural map* $j\colon X\to X''$, $x\mapsto g_x$, is linear. It will be shown that $\|g_x\|=\|x\|$ holds with equality, so that j is norm-preserving, and in particular injective.

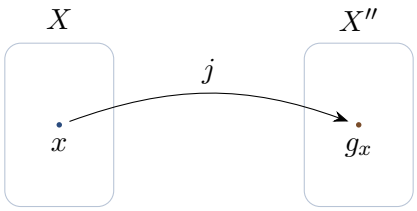


Figure 11.1. *The natural map j sends $x\in X$ to the functional $g_x\in X''$, which evaluates every $f\in X'$ at x: $g_x(f)=f(x)$.*

Definition 11.6 (Reflexive space)**.** A normed space $(X,\|\cdot\|)$ is *reflexive* if the natural map $j\colon X\to X''$ is bijective, that is, if every element of X'' can be written as g_x for some $x\in X$. In the sense of Remark 11.1 this is written $((X,\|\cdot\|)')'=(X,\|\cdot\|)$.

Proposition 11.7. Every reflexive space is a Banach space.

Proof. The proof rests on the norm equality $\|g_x\| = \|x\|$ announced in Remark 11.5. The bidual $X'' = \mathcal{B}(X', \mathbb{K})$ is a Banach space by Corollary 9.6, applied to the normed space X'. Let (x_n) be a Cauchy sequence in X. Since $\|g_{x_n} - g_{x_m}\| = \|j(x_n - x_m)\| = \|x_n - x_m\|$, the sequence (g_{x_n}) is Cauchy in X'' and converges to some $\varphi \in X''$. As X is reflexive, $\varphi = g_x$ for some $x \in X$, and then $\|x_n - x\| = \|g_{x_n} - g_x\| \to 0$. So X is complete. ∎

Example 11.8. ℓ_p is reflexive for $1 < p < \infty$; ℓ_1 and ℓ_∞ are not.

11.2. The dual of $C[a, b]$

Definition 11.9 (Bounded variation)**.** A function $w\colon [a, b] \to \mathbb{K}$ is *of bounded variation* if

$$\operatorname{Var}(w) = \sup \sum_{j=1}^{n} |w(t_j) - w(t_{j-1})| < \infty,$$

where the supremum is taken over all partitions $a = t_0 < t_1 < \cdots < t_n = b$ of $[a, b]$. Such functions carry the norm $\|w\| = |w(a)| + \operatorname{Var}(w)$.

Theorem 11.10 (Representation of $(C[a, b])'$)**.** For every $f \in (C[a, b])'$ there is a w of bounded variation on $[a, b]$ such that

$$f(g) = \int_a^b g(t)\mathrm{d}w(t) \qquad \text{for all } g \in C[a, b].$$

The function w is not unique. If $w \in C^1[a, b]$, then $f(g) = \int_a^b g(t)\, w'(t)\mathrm{d}t$.

12. Inner product spaces

In three-dimensional Euclidean space there is a dot product, also called inner or scalar product. Inner product spaces generalize this concept.

Definition 12.1 (Inner product space)**.** Let X be a vector space over $\mathbb{K}$. A map $\langle \cdot, \cdot \rangle\colon X \times X \to \mathbb{K}$ is an *inner product*, and $(X, \langle \cdot, \cdot \rangle)$ is an *inner product space*, or *pre-Hilbert space*, if for all $x, y, z \in X$ and $\alpha \in \mathbb{K}$:

1. $\langle x + y, z \rangle = \langle x, z \rangle + \langle y, z \rangle$ and $\langle \alpha x, y \rangle = \alpha \langle x, y \rangle$, that is, $\langle \cdot, \cdot \rangle$ is linear in the first variable;
2. $\langle x, y \rangle = \overline{\langle y, x \rangle}$, conjugate symmetry;
3. $\langle x, x \rangle \geq 0$, and $\langle x, x \rangle = 0$ if and only if $x = 0$, positive definiteness.

For $\mathbb{K} = \mathbb{R}$ the conjugation is the identity.

Remark 12.2. An inner product is conjugate linear in its second variable: $\langle x, y + z \rangle = \langle x, y \rangle + \langle x, z \rangle$ and $\langle x, \alpha y \rangle = \bar{\alpha} \langle x, y \rangle$, by conjugate symmetry and linearity in the first variable.

Definition 12.3 (Hilbert space)**.** A *Hilbert space* is an inner product space that is complete with respect to the norm $\|x\| = \sqrt{\langle x, x \rangle}$, $x \in X$.

Proposition 12.4. $\|x\| = \sqrt{\langle x, x\rangle}$ is a norm on X, the *induced norm*. Its metric $d(x, y) = \|x - y\| = \sqrt{\langle x - y, x - y\rangle}$, as in Definition 2.1, is the *induced metric*.

Definition 12.5 (Orthogonality)**.** Vectors $x, y \in X$ are *orthogonal*, written $x \perp y$, if $\langle x, y\rangle = 0$.

Proposition 12.6 (Parallelogram identity)**.** For all $x, y \in X$, $\|x+y\|^2+\|x-y\|^2 = 2\|x\|^2+2\|y\|^2$.

Proof. Expanding by linearity in each variable,

$$\|x \pm y\|^2 = \langle x \pm y, x \pm y\rangle = \|x\|^2 \pm \langle x, y\rangle \pm \langle y, x\rangle + \|y\|^2.$$

Summing the two cases, the mixed terms cancel, and the sum is $2\|x\|^2 + 2\|y\|^2$. ∎

Theorem 12.7. A norm is induced by an inner product if and only if it satisfies the parallelogram identity of Proposition 12.6.

Example 12.8. $\ell_2 = \left\{x = (x_1, x_2, \dots) \,\middle|\, \sum_{n=1}^{\infty} |x_n|^2 < \infty\right\}$ is an inner product space with

$$\langle x, y\rangle = \sum_{n=1}^{\infty} x_n \overline{y_n},$$

the series converging absolutely by Hölder's inequality, Theorem 2.6, with $p = q = 2$.

Example 12.9. Let $L_2[a, b]$ be defined as $L_2[0, 1]$ in Definition 2.8, with $[a, b]$ in place of $[0, 1]$. It is an inner product space with $\langle f, g\rangle = \int_a^b f\bar{g}$.

Theorem 12.10 (Cauchy–Schwarz inequality)**.** For all x, y in an inner product space X,

$$|\langle x, y\rangle| \le \|x\|\,\|y\|.$$

Proof. If $y = 0$ both sides vanish, so let $y \neq 0$. For every $\lambda \in \mathbb{K}$, by positivity and Remark 12.2,

$$0 \le \|x - \lambda y\|^2 = \|x\|^2 - \bar{\lambda}\langle x, y\rangle - \lambda\langle y, x\rangle + |\lambda|^2\|y\|^2.$$

Choose $\lambda = \langle x, y\rangle / \|y\|^2$. Then each of the last three terms has absolute value $|\langle x, y\rangle|^2/\|y\|^2$, the first two with a minus sign, so

$$0 \le \|x\|^2 - \frac{|\langle x, y\rangle|^2}{\|y\|^2}.$$

Multiplying by $\|y\|^2$ and taking square roots gives the claim. ∎

Theorem 12.11 (Triangle inequality for the induced norm)**.** For all x, y in an inner product space X, $\|x + y\| \le \|x\| + \|y\|$.

Proof. By conjugate symmetry $\langle x, y\rangle + \langle y, x\rangle = 2\,\mathrm{Re}\langle x, y\rangle$, so Theorem 12.10 gives

$$\|x + y\|^2 = \|x\|^2 + 2\,\mathrm{Re}\langle x, y\rangle + \|y\|^2 \le \|x\|^2 + 2\|x\|\,\|y\| + \|y\|^2 = \big(\|x\| + \|y\|\big)^2.$$

Taking square roots gives the claim, which is the triangle inequality of Proposition 12.4. ∎

13. Best approximation and orthogonal complements

Definition 13.1 (Convex set)**.** A set M in a vector space is *convex* if $tx + (1-t)y \in M$ for all $x, y \in M$ and $t \in [0,1]$.

Theorem 13.2 (Minimizing vector)**.** Let X be an inner product space and $M \subset X$ a nonempty, convex and complete subset. Then for every $x \in X$ there is a unique $y \in M$ with

$$\|x - y\| = \operatorname{dist}(x, M),$$

where $\operatorname{dist}(x, M) = \inf\{\|x - m\| : m \in M\}$, as in Lemma 5.7.

Remark 13.3. If X is complete, a subset of X is complete exactly when it is closed, so in a Hilbert space the hypothesis reads: M is nonempty, convex and closed.

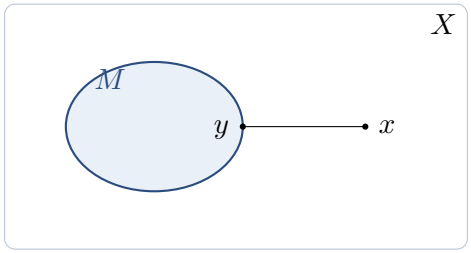


Figure 13.1. *The closest point $y \in M$ to a point x outside the convex set M.*

Proof of Theorem 13.2. As M is nonempty, $\delta = \operatorname{dist}(x, M)$ is finite, and by the definition of the infimum there are $y_n \in M$ with $\|y_n - x\| \to \delta$. We show that (y_n) is Cauchy. Apply the parallelogram identity, Proposition 12.6, to $y_n - x$ and $y_m - x$:

$$\|y_n + y_m - 2x\|^2 + \|y_n - y_m\|^2 = 2\|y_n - x\|^2 + 2\|y_m - x\|^2.$$

By convexity $\frac{y_n + y_m}{2} \in M$, so $\|y_n + y_m - 2x\|^2 = 4\left\|\frac{y_n + y_m}{2} - x\right\|^2 \geq 4\delta^2$, and therefore

$$\|y_n - y_m\|^2 \leq 2\|y_n - x\|^2 + 2\|y_m - x\|^2 - 4\delta^2 \xrightarrow[n,m\to\infty]{} 2\delta^2 + 2\delta^2 - 4\delta^2 = 0.$$

So (y_n) is Cauchy, and as M is complete, $y = \lim_{n\to\infty} y_n$ exists and lies in M. Then $\delta \leq \|x - y\|$ because $y \in M$, and $\|x - y\| \leq \|x - y_n\| + \|y_n - y\| \to \delta$, so $\|x - y\| = \delta$.

The point y is unique. Let $y_1, y_2 \in M$ with $\|y_1 - x\| = \|y_2 - x\| = \delta$. The same estimate, with y_1 and y_2 in place of y_n and y_m, gives

$$\|y_1 - y_2\|^2 \leq 2\|y_1 - x\|^2 + 2\|y_2 - x\|^2 - 4\delta^2 = 2\delta^2 + 2\delta^2 - 4\delta^2 = 0,$$

so $y_1 = y_2$. ∎

Definition 13.4. A vector $x \in X$ is *orthogonal* to a set $M \subset X$, written $x \perp M$, if $x \perp m$ for every $m \in M$.

Lemma 13.5 (Orthogonality)**.** Let M be a complete subspace of an inner product space X, let $x \in X$, and let $y \in M$ be the closest point of M to x, which exists and is unique by Theorem 13.2, a subspace being convex. Then $x - y \perp M$.

Proof. Let $y_1 \in M$; we show $\langle y_1, y - x\rangle = 0$. If $y_1 = 0$ this holds, so let $y_1 \neq 0$. For $\alpha \in \mathbb{K}$ the vector $y + \alpha y_1$ lies in the subspace M, so by the minimality of y,

$$\begin{aligned}\|y - x\|^2 \leq \|y + \alpha y_1 - x\|^2 &= \|y - x\|^2 + \alpha\langle y_1, y - x\rangle + \bar{\alpha}\langle y - x, y_1\rangle + \alpha\bar{\alpha}\|y_1\|^2 \\ &= \|y - x\|^2 + \alpha\big(\langle y_1, y - x\rangle + \bar{\alpha}\|y_1\|^2\big) + \bar{\alpha}\langle y - x, y_1\rangle.\end{aligned}$$

Choose $\bar{\alpha} = -\langle y_1, y - x\rangle / \|y_1\|^2$. The bracket vanishes, and

$$\|y - x\|^2 \leq \|y - x\|^2 - \frac{\langle y_1, y - x\rangle}{\|y_1\|^2}\langle y - x, y_1\rangle = \|y - x\|^2 - \frac{|\langle y_1, y - x\rangle|^2}{\|y_1\|^2}.$$

Hence $|\langle y_1, y - x\rangle|^2 \leq 0$, that is $\langle y_1, y - x\rangle = 0$. By conjugate symmetry and linearity, $\langle x - y, y_1\rangle = 0$. As $y_1 \in M$ was arbitrary, $x - y \perp M$. ∎

13.1. Complementary subspaces and projections

Definition 13.6 (Complementary subspaces)**.** Let X be a vector space. Subspaces $M, N \subset X$ are *complementary* if $M \cap N = \{0\}$ and $\operatorname{span}(M \cup N) = X$.

Remark 13.7. For any two subspaces M and N, $\operatorname{span}(M \cup N) = M + N = \{u + w \mid u \in M, w \in N\}$.

Lemma 13.8. Subspaces M and N are complementary in X if and only if every $x \in X$ has a unique decomposition $x = x_M + x_N$ with $x_M \in M$ and $x_N \in N$.

Proof. Assume M and N are complementary. Existence is immediate from $M + N = X$. If $x = u_1 + w_1 = u_2 + w_2$ with $u_1, u_2 \in M$ and $w_1, w_2 \in N$, then $u_1 - u_2 = w_2 - w_1 \in M \cap N$, so $u_1 - u_2 = w_2 - w_1 = 0$, and the decomposition is unique.

Conversely, $M + N = X$ is immediate from existence. If $x \in M \cap N$, then $x = x + 0 = 0 + x$ are two decompositions, with the first summand in M and the second in N, so by uniqueness $x = 0$, and $M \cap N = \{0\}$. ∎

Definition 13.9 (Projection)**.** Let M and N be complementary in X. The map $P\colon X \to X$, $x \mapsto x_M$, given by Lemma 13.8, is the *projection onto M along N*. A linear map $P\colon X \to X$ is a *projection* if it is the projection onto some subspace along a complementary one.

Remark 13.10. The projection onto M along N is linear: for $x, x' \in X$ and $\alpha \in \mathbb{K}$, the sum $(x_M + \alpha x'_M) + (x_N + \alpha x'_N)$ is a decomposition of $x + \alpha x'$, so by uniqueness $P(x + \alpha x') = Px + \alpha Px'$.

Proposition 13.11. Let P be the projection onto M along N. Then:

1. $Pu = u$ for every $u \in M$, since $u = u + 0$ is its decomposition;
2. $\operatorname{im} P \subset M$, since $Px = x_M \in M$;
3. $\operatorname{im} P = M$, by item 2 and by item 1, which writes each $u \in M$ as Pu;
4. $P^2 = P$, since $Px \in M$, so $P(Px) = Px$ by item 1;
5. $\ker P = N$, since $Px = 0$ exactly when $x_M = 0$, that is, when $x = x_N \in N$;
6. $I - P$ is the projection onto N along M, since $(I - P)x = x - x_M = x_N$.

Proposition 13.12. A linear map $P\colon X \to X$ is a projection if and only if $P^2 = P$, that is, P is *idempotent*.

Proof. If P is a projection, $P^2 = P$ by Proposition 13.11.

Conversely, let $P^2 = P$. The image and the kernel of the linear map P are subspaces. For $x \in X$ write $x = Px + (x - Px)$. Here $Px \in \operatorname{im} P$, and $x - Px \in \ker P$ because $P(x - Px) = Px - P^2x = 0$. So $\operatorname{im} P + \ker P = X$. If $x \in \operatorname{im} P \cap \ker P$, then $Px = 0$ and $x = Pz$ for some $z \in X$, so $x = Pz = P^2z = Px = 0$. Thus $\operatorname{im} P$ and $\ker P$ are complementary, and since the $\operatorname{im} P$ component of x is Px, P is the projection onto $\operatorname{im} P$ along $\ker P$. ∎

13.2. Orthogonal complements

Definition 13.13 (Orthogonal complement)**.** Let X be an inner product space and $M \subset X$ a subspace. Its *orthogonal complement* is $M^\perp = \{x \in X \mid x \perp M\}$.

Proposition 13.14. Let M and N be subspaces of an inner product space X. Then:

1. $M^\perp$ is a subspace, since $\langle \alpha x + \beta x', m\rangle = \alpha\langle x, m\rangle + \beta\langle x', m\rangle$;
2. $M \subset N$ implies $N^\perp \subset M^\perp$, since a vector orthogonal to every element of N is orthogonal to every element of M;
3. $M \cap M^\perp = \{0\}$, since x in both has $\langle x, x\rangle = 0$;
4. $M \subset (M^\perp)^\perp$, since $\langle u, x\rangle = \overline{\langle x, u\rangle} = 0$ for $u \in M$ and $x \in M^\perp$.

Counterexample 13.15. In general $M + M^\perp \neq X$ and $(M^\perp)^\perp \neq M$.

Proof. In ℓ_2 let M consist of the sequences with only finitely many nonzero terms, and let e_i be the standard basis vectors. Then $M \neq \ell_2$, since $(2^{-n})_n \in \ell_2$ has infinitely many nonzero terms. If $x \in M^\perp$, then $0 = \langle x, e_1\rangle = x_1$, $0 = \langle x, e_2\rangle = x_2$, and so on, so $x = 0$ and $M^\perp = \{0\}$. Hence $M + M^\perp = M \neq \ell_2$ and $(M^\perp)^\perp = \{0\}^\perp = \ell_2 \neq M$. ∎

Proposition 13.16. Let M be a complete subspace of an inner product space X. Then $M + M^\perp = X$ and $(M^\perp)^\perp = M$.

Proof. M is nonempty, as $0 \in M$, and convex, being a subspace. For $x \in X$ let $x_M \in M$ be the closest point of M to x, by Theorem 13.2; then $x - x_M \in M^\perp$ by Lemma 13.5. So $x = x_M + (x - x_M)$, and $M + M^\perp = X$.

By Proposition 13.14 it remains to check $(M^\perp)^\perp \subset M$. Let $x \in (M^\perp)^\perp$, and take x_M as above, so that $x - x_M \in M^\perp$. Then $\langle x, x - x_M\rangle = 0$, because $x \in (M^\perp)^\perp$, and $\langle x_M, x - x_M\rangle = 0$, because $x_M \in M$. Subtracting,

$$\|x - x_M\|^2 = \langle x - x_M, x - x_M\rangle = \langle x, x - x_M\rangle - \langle x_M, x - x_M\rangle = 0,$$

so $x = x_M \in M$. ∎

14. The dual of a Hilbert space

Theorem 14.1 (Riesz representation)**.** Let $(H, \langle\cdot,\cdot\rangle)$ be a Hilbert space and $f \in H'$. Then there is a unique $y \in H$ such that $f(x) = \langle x, y\rangle$ for all $x \in H$. Moreover, for every $y \in H$ the map $x \mapsto \langle x, y\rangle$ is in H'. So the elements of H' are precisely the functionals of the form $\langle\cdot, y\rangle$, $y \in H$.

Proof. Let $y \in H$. Then $\langle\cdot, y\rangle\colon H \to \mathbb{K}$ is linear, and $|\langle x, y\rangle| \le \|x\|\,\|y\|$ for every $x \in H$ by Theorem 12.10, so it is bounded, with $\|\langle\cdot, y\rangle\| \le \|y\|$. For $y = 0$ equality holds; for $y \neq 0$ the unit vector $x = y/\|y\|$ gives $|\langle x, y\rangle| = \|y\|$. Hence $\|\langle\cdot, y\rangle\| = \|y\|$, and $\langle\cdot, y\rangle \in H'$.

Now let $f \in H'$. If $f = 0$, take $y = 0$. Let $f \neq 0$ and $M = \ker f \neq H$. M is a closed subspace, being the preimage of the closed set $\{0\} \subset \mathbb{K}$ under the map f, which is continuous by Theorem 9.1. H is complete and M is closed, so M is complete as well, and by Propositions 13.16 and 13.14 the subspaces M and $M^\perp$ are complementary in H. Since $M \neq H = M + M^\perp$, there is a $z \in M^\perp$ with $z \neq 0$. Then $z \notin M$, as $M \cap M^\perp = \{0\}$, so $f(z) \neq 0$.

We claim $M^\perp = \mathbb{K}z$. The inclusion $\mathbb{K}z \subset M^\perp$ holds because $M^\perp$ is a subspace. If $\tilde z \in M^\perp$, then

$$f\Big(\tilde z - \frac{f(\tilde z)}{f(z)}\, z\Big) = f(\tilde z) - \frac{f(\tilde z)}{f(z)}\, f(z) = 0,$$

so $\tilde z - \frac{f(\tilde z)}{f(z)} z \in M \cap M^\perp = \{0\}$, and $\tilde z = \frac{f(\tilde z)}{f(z)} z$.

We claim $f(x) = \frac{f(z)}{\|z\|^2}\langle x, z\rangle$ for all $x \in H$. Let $x = x_M + (x - x_M)$ with $x_M \in M$ and $x - x_M \in M^\perp$; by the previous claim $x - x_M = \lambda z$ for some $\lambda \in \mathbb{K}$. Then

$$f(x) = f(x_M) + f(x - x_M) = 0 + \lambda f(z),$$

and, as $x_M \perp z$, $\langle x, z\rangle = \langle x_M, z\rangle + \lambda\|z\|^2 = \lambda\|z\|^2$, so $\lambda = \langle x, z\rangle/\|z\|^2$. Moving the scalar into the second variable, which is conjugate linear by Remark 12.2,

$$f(x) = \frac{f(z)}{\|z\|^2}\langle x, z\rangle = \Big\langle x, \frac{\overline{f(z)}}{\|z\|^2}\, z\Big\rangle = \langle x, y\rangle, \qquad y = \frac{\overline{f(z)}}{\|z\|^2}\, z.$$

Finally, y is unique. If $\langle x, y\rangle = \langle x, y'\rangle$ for all $x \in H$, then $x = y - y'$ gives $\|y - y'\|^2 = 0$, so $y = y'$. ∎